whether any nodes not included in the subgraph are important for task behavior. Additional ad hoc validation methodologies also exist. For example, Wang et al. (2023) generate adversarial examples for their task based on their mechanistic understanding, while Shi et al. (2024) devise a suite of formal statistical hypothesis tests for circuit efficacy.

This circuit discovery procedure has yielded valuable insight, but falls short using current methods. The pipeline has several issues:

**Task definition is concept-based**. Defining circuits has thus far been with respect to tasks defined by humans. Miller et al. (2024) demonstrate that the within-task variance of model performance across the distribution of data points in a task is large, implying that the circuit provides a good approximation of the average case performance on the dataset, but a poor one for any individual data point. This suggests that the process of first selecting a task and then studying how the model performs it may not be an effective approach to achieve "reverse engineering" -style interpretability. Thus, it might be worth learning the task decomposition instead (Haani et al., 2024).

**Network decomposition methods are flawed**. Perhaps most importantly, prior circuit discovery work has attempted to decompose models in either architectural bases (Wang et al., 2023; Conmy et al., 2023) or sparse autoencoder latents (Marks et al., 2024; Huben et al., 2024), which are imperfect ways to decompose neural networks for mechanistic interpretability (Section 2.1.2). Future work could locate circuits in improved decompositions or simultaneously learn both network decompositions and circuits.

**Circuit faithfulness is low**. Simple early circuits were found to be unfaithful (Chan et al., 2022b; 2023). Miller et al. (2024) show that existing measures of faithfulness depend on the causal intervention implementation used, and further demonstrate that such metrics are misleading when applied to several complex end-to-end circuits. Makelov et al. (2023) argue that subspace activation patching via distributed alignment search may lead to interpretability illusion mechanisms, although these findings are contested by Wu et al. (2024c).

**Scalable methods are only approximate**. Identifying relevant components through individual interventions is costly when there are many components. Attribution patching Syed et al. (2024) was designed to identify potential relevant candidates for further testing through intervention, which becomes more important as the number of components expands significantly through sparse dictionary learning (Marks et al., 2024). However, attribution patching uses gradients, which only yield a first-order approximation of the effect of ablating components (Wu et al., 2024c; Molchanov et al., 2017), leaving it unclear whether this method and any improvements on it (Kramár et al., 2024) produce adequate approximations.

**Circuit discovery algorithms struggle with backup and negative behavior**. Additional challenges for circuit analysis arise from the effects of "backup" and "negative" behavior (Wang et al., 2023; McGrath et al., 2023; McDougall et al., 2024), which actively suppress task performance and are thus not captured by maximizing task performance metrics. Despite this, they remain important factors to consider; Mueller (2024) provides further discussion of these issues.

**Streetlight Interpretability**: The tasks studied so far have been deliberately selected to be simple to define and study mechanistically (Wang et al., 2023). This gives a misleading impression of the level of difficulty involved in implementing circuit discovery for any arbitrary task that a network implements. Indeed, attempts to study arbitrary circuits have proceeded less successfully (Nanda et al., 2023b).

Solving issues with current mechanistic interpretability pipelines remains an open challenge that promises significant benefit. Upon establishing reasonable procedures, automating the overall pipeline will become more feasible. However, some individual steps in mechanistic interpretability can already be fruitfully automated, as discussed in the next section (Section 2.4).

### 2.4 Automating steps in mechanistic interpretability research

Historically, mechanistic interpretability research has required considerable manual researcher effort, though it typically studies models that are smaller than those at the frontier. To make interpretability useful for downstream use cases, scalable approaches are crucial. In this section, we discuss *automated interpretability*

methods. We will explore previous cases where manual tasks in mechanistic interpretability have been successfully automated and address open problems in further automation.

**Automating feature description and validation**. A task that is amenable to automation is 'describing the functional role of model components' (Section 2.1.3). With the increasing sophistication of language models, researchers have generated descriptions of the functional role of neurons in image models (Hernandez et al., 2022), neurons in language models (Bills et al., 2023), and sparse autoencoder latents in language models (Huben et al., 2024; Bricken, 2023; Juang et al., 2024) using highly activating data set examples. These interpretations are validated by assessing how effectively a human or model can use them to predict the activation of a feature in a given data set example, or predict where a feature is active within a single image or text excerpt. The success of these predictions can be used as a quantitative measurement of 'interpretability'. This was previously used to measure progress toward a decomposition method that carves networks at the joints of their generalization structure, assuming that such a decomposition would be maximally interpretable (Section 2.1.2). While imperfect, these methods for interpretation hypothesis generation and validation might be improved by automating the generation of inputs to test the interpretation hypotheses by ensuring that generations activate the interpreted feature (Huang et al., 2023), or defining more rigorous statistical tests (Bloom & Lin, 2024b). Automatic component labeling could expand in the future to include descriptions of feature effects, relationships between features (Bussman et al., 2024), or how components interact during runtime produce behavior.

**Automating circuit discovery pipelines**. An approach called Automated Circuit DisCovery' (ACDC) automates part of the pipeline discussed in Section 2.3 to identify computational subgraphs involved in particular tasks (Conmy et al., 2023). Several works have since improved upon and accelerated this process (Syed et al., 2024; Kramár et al., 2024; Marks et al., 2024). Note that ACDC-like approaches in general assist in identifying relevant subgraphs for a pre-defined task, but do not automate important subsequent steps, such as describing the functional role of subgraph components.

While significant progress has been made toward automating steps of mechanistic interpretability pipelines, fully automating current pipelines would not yield satisfactory explanations of model behavior[6]. Further methodological progress is required for fully automated neural network interpretability to be capable of generating the quality of interpretations necessary to achieve our goals.

[6]For one attempt at this using leading decomposition and description methods, see Marks et al. (2024)

## 3 Open problems in applications of mechanistic interpretability

Ultimately, we need mechanistic interpretability methods that enable us to solve concrete scientific and engineering problems (Figure **6**). While predicting the impact of fundamental science in advance is difficult, having concrete goals in mind during research is usually beneficial. We want mechanistic interpretability methods to help us achieve various outcomes, such as monitoring and auditing AI systems more effectively (Section 3.2), controlling of AI system behavior more precisely (Section 3.2.2), predicting AI system outcomes more accurately (Section 3.3), enhancing AI system capabilities (Section 3.4), and extracting knowledge from AI systems (Section 3.5). We should also anticipate that mechanistic interpretability will uncover "unknown problems" present in systems, revealing that the true realm of challenges and possibilities is greater than what we currently perceive it to be.

As highlighted in the previous section, progress in mechanistic interpretability methods is multifaceted. Each axis of methodological advancement leads to varying degrees of progress toward different goals. Before we discuss open questions in its applications, we identify distinct axes of methodological progress that lead to different amounts of progress toward different goals.

### 3.1 Axes of mechanistic interpretability progress.

**Decomposition vs. description of network components**: Improvements in network decomposition versus component description methods offer varying benefits for different goals. Decomposition methods vary in their efficacy at carving networks at the joints of their generalization structure, while description methods can yield descriptions that vary in depth. Deeper descriptions of a component are typically more causal or mechanistic, whereas shallower descriptions may rely more on correlations and only connect to inputs or outputs without referencing intermediate causes or effects (Figure **5**. Deeper descriptions thus attempt to explain more about how the component interacts with other components within the network's algorithm. Certain goals can be achieved with minimal or no progress in decomposition or description, while others may demand substantial progress.

**Extent of network decomposition or description**: The extent of network decomposition or description needed may vary depending on the goal. Certain goals only require an understanding of specific network components (such as an individual features or a circuit), while others might require enumerating or understanding of larger circuits or the entire model (as in 'enumerative safety').

**Extent of task distribution analyzed**: The scope of task distribution analysis also depends on the intended goal. For instance, monitoring a model for a single kind of behavior might only require decomposing or understanding the model only over a narrow task distribution, while others, such as formal verification, might demand understanding over the entire distribution of tasks.

**Mechanistic understanding post vs. during training**: Understanding the mechanisms of a fixed model might suffice for some goals, but more ambitious goals might require an understanding of not only the models' mechanisms, but also how they change during the learning process.

In this section, we'll discuss how mechanistic interpretability has been used or could be leveraged to further the field's various goals. We'll assess the progress made thus far, and identify the advancements along different axes of methodological progress that will be most crucial to success.

### 3.2 Using mechanistic interpretability for better monitoring and auditing of AI systems for potentially unsafe cognition

#### 3.2.1 Mechanistic interpretability-based evaluations could help us detect unsafe or unethical AI cognition

Currently, we rely on "black box" evaluations to understand a model's capabilities, but studying input-output behavior alone may not reveal all dangerous behaviors. Such behaviors include deceiving users (Park et al., 2023b; Ward et al., 2023; Scheurer et al., 2024; Meinke et al., 2024); for instance, by intentionally underperforming on evaluations ("sandbagging", van der Weij et al. (2024)), leveraging situational awareness